

GRADE 10 STE

Digital and Analog Input Systems



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Digital Input

Binary States and Logic Control

| Definition of Terms



Digital Input (DI)

A circuit that receives binary signals (1 or 0) from external sensors or switches to communicate with a controller.



Binary State

The two possible conditions of a digital signal: ON (High Voltage) or OFF (Low Voltage/Ground).



Logic Levels

Specific voltage thresholds that represent 1 and 0 (e.g., 5V for High, 0V for Low).

| Different Types



Mechanical Switches

Physical devices like push buttons and selector switches that open or close a circuit.



Digital Sensors

Proximity, motion, or infrared sensors that output high/low signals based on detection.



Encoders

Devices that convert rotational or linear motion into a sequence of digital pulses.

| Functions



Status Monitoring

Digital inputs are primarily used to check the status of devices—whether a safety gate is closed, a motor is running, or a button is pressed.



Logic Triggering

They act as the "conditions" in automation. For example: "IF the button is pressed (1), THEN start the conveyor belt."

| Examples & Images

 Push Button Switch

Push Buttons

Standard momentary switches for user commands.



Limit Switches

Detects physical contact in machines.



Proximity Sensors

Detects objects without physical contact.

Analog Input

Continuous Signals and Variable Data

| Definition of Terms

Analog Input

A continuous electrical signal that represents a physical quantity over a defined range.

Resolution

The precision of the input, determined by the number of bits used to represent the continuous signal (e.g., 10-bit or 12-bit).

Signal Range

Common industrial standards include 0-10 Volts or 4-20 milliamperes (mA).

| Different Types



Voltage Inputs

Signals ranging from 0V to 5V or 10V, commonly used for position and light sensing.



Current Loops

The 4-20mA standard used in industrial sensors for long-distance stability.



Resistance

Sensors that vary in electrical resistance, such as thermistors and potentiometers.

| Functions



Precise Measurement

Allows measuring variable physical quantities like specific temperature degrees, pressure levels, or flow rates rather than just ON/OFF.



Intensity Tracking

Enables the system to respond to changes in magnitude, such as adjusting light brightness or fan speed proportionally.

| Examples & Images

 Potentiometer

Potentiometers

Rotating knobs for volume or speed control.



Temp Sensors

Thermocouples used to monitor heat levels.

 Joystick Module

Joysticks

Analog sticks for multi-axis movement control.

Questions?

Thank you for listening!

Grade 10 STE | Group Presentation

| Image Sources



<https://cdn.sparkfun.com/r/600-600/assets/parts/1/2/2/1/8/14460-01.jpg>

Source: news.sparkfun.com



<https://vva-industrial.net/wp-content/uploads/2025/08/Sensor-Limit-Switch-1.png>

Source: vva-industrial.net



<https://lorentzzi.com/wp-content/uploads/2023/03/LM14-inductive-sensor-1.jpg>

Source: lorentzzi.com



<https://www.chinadaier.com/wp-content/uploads/2024/04/KN-1316-Aluminum-Rotary-Electronic-Control-Potentiometer-Knob-Application.jpg.webp>

Source: www.chinadaier.com



Thumbnail

for

https://cdn11.bigcommerce.com/s-sgprcd6/images/stencil/1193x795/uploaded_images/temperature-sensors.jpg?t=1656850634

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